Fairness / Bias Report

1: Definition of Groups

- **Sex:** Evaluates whether predictive accuracy differs systematically between male and female populations.
- **Race/Ethnicity:** Assesses potential disparities in model performance across racial or ethnic groups, given known structural inequalities in health systems.
- **Age:** Examines performance variation across age strata. While epidemiological differences are expected, disproportionate predictive error may indicate model sensitivity to demographic composition.
- **Region:** Evaluates territorial disparities in predictive performance. Regional heterogeneity in infrastructure and reporting practices may affect model reliability.
- **Hospital Size**: Examines model's predictive capabilities based on hospital's size.

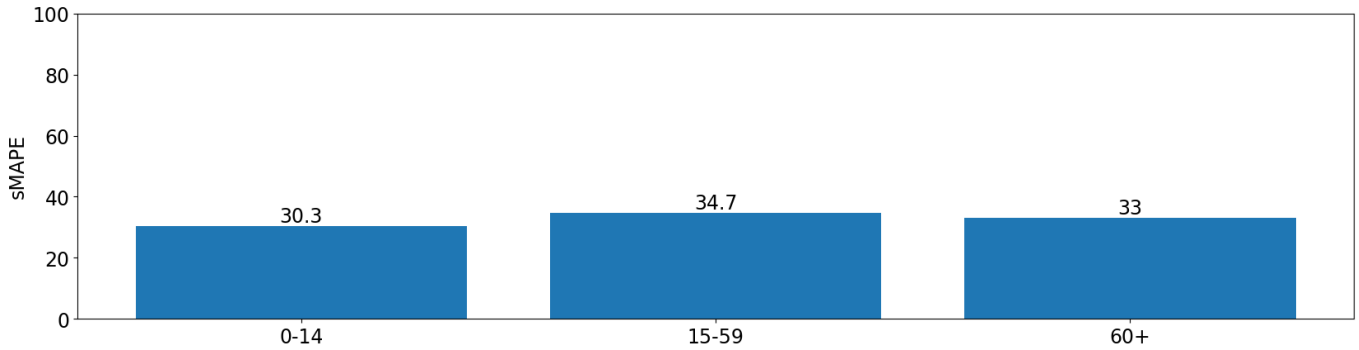
2: Metric Selection

Fairness comparisons were conducted using the Symmetric Mean Absolute Percentage Error (SMAPE). Percentage-based metrics were preferred because they allow relative comparison across groups with different baseline incidence levels.

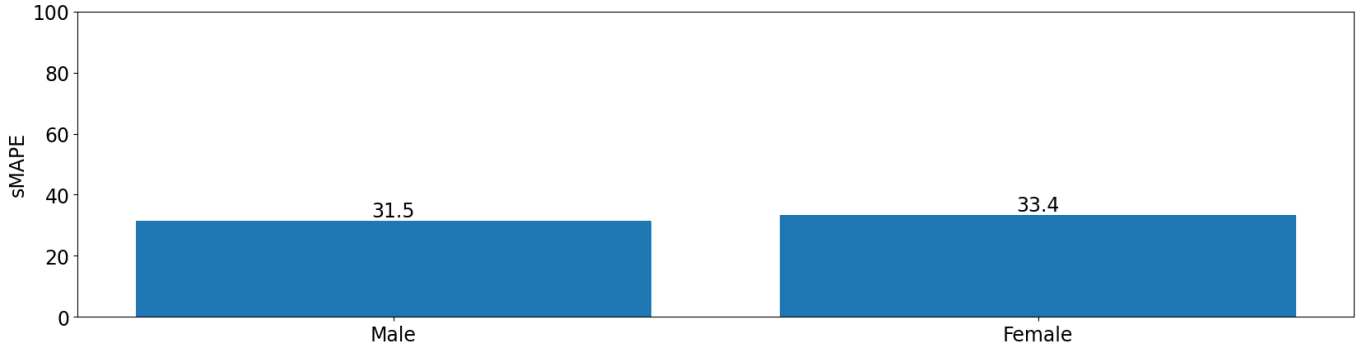
Absolute error measures such as MAE or RMSE are scale-dependent and may not be directly comparable when hospitalization volumes differ substantially between groups. SMAPE provides a normalized measure of predictive deviation, facilitating cross-group disparity assessment.

Base results:

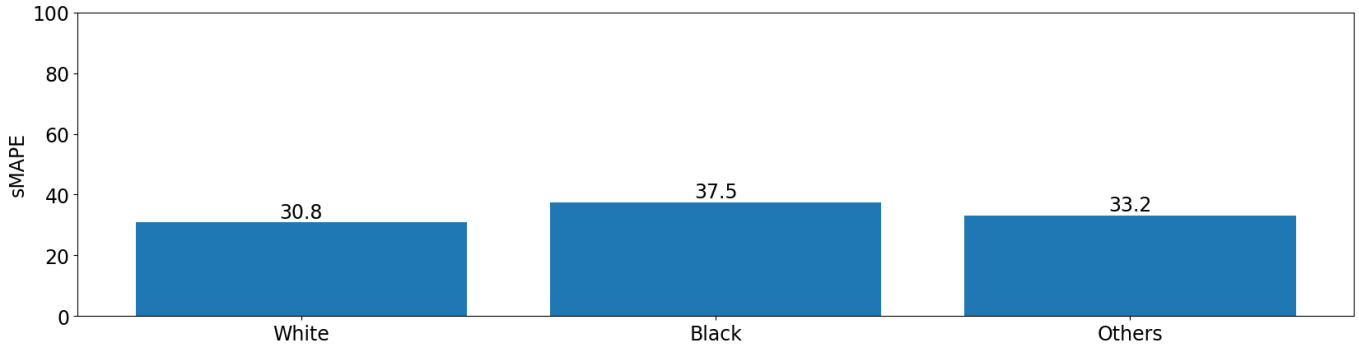
- Age:



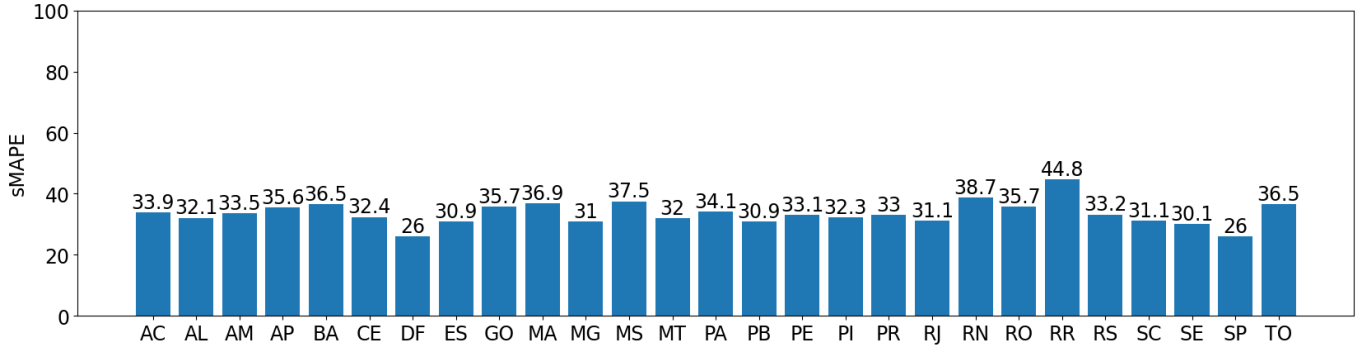
- Sex:



- Race:



- Region:



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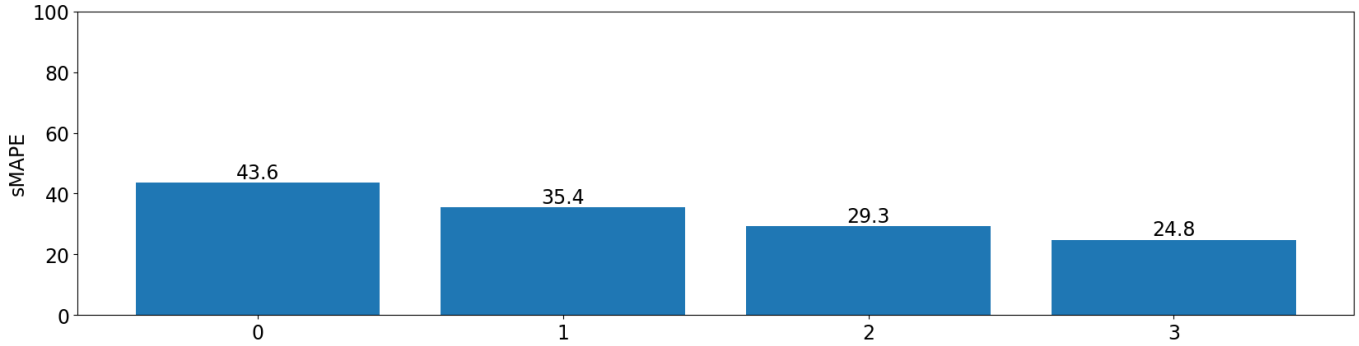


3: First Evaluation

Analyses from the results reveal a low disparity in model performance when looking at different groups based on age, sex and race. On the other hand, efficiency is widely different when looking through different regions and hospital sizes, both portraying a difference of 18.8% when comparing best and worst cases.

An attempt was done at bias correction through resampling (more on the next section), though only region based resampling led to a positive outcome. Every attempt at resampling data based on hospital size would at least triple error metrics and drastically increase disparity across all groups. For this reason, only region received bias mitigation treatment.

4: Bias mitigation

To address the identified regional disparity, we implemented a mitigation strategy based on data resampling. While tools like Fairlearn offer algorithmic constraints, resampling was selected as an initial intervention due to its transparency and minimal architectural modification requirements.

We employed SMOTE-NC, an adaptation of the Synthetic Minority Over-sampling Technique for mixed-data types, to over-sample underrepresented regions. Over-sampling was preferred to preserve the overall training data volume. Resampling was applied exclusively to the training partition after the split. Synthetic samples were generated within the training data only, preserving the prospective evaluation design and preventing temporal leakage. A fixed randomized seed was used, with value 18979.

- **Base sample:**

Age:

```
          AGE
 --------+---------------------
 group   |        Total
 --------+---------------------
 60+     |        32461
 15-59   |        4145
 0-14    |        19644
```

Sex:

```
          SEX
 --------+---------------------
 group   |        Total
 --------+---------------------
 Male    |        34056
 Female  |        22194
```

Race:

```
          RACE
 --------+---------------------
 group   |        Total
 --------+---------------------
 White   |        21003
 Black   |        241
 Others  |        35006
```

Region:

```
          REGION
--------+---------------------
group   |        Total
--------+---------------------
MA      |        3127
BA      |        4594
CE      |        2908
RN      |        979
PR      |        3705
MG      |        6904
AC      |        243
SP      |        8027
RR      |        147
PA      |        2678
PI      |        1238
SC      |        2393
AL      |        480
SE      |        360
AM      |        963
GO      |        2119
PE      |        2442
MS      |        1136
TO      |        467
RO      |        761
DF      |        331
MT      |        1365
ES      |        913
PB      |        1145
RS      |        4011
AP      |        170
RJ      |        2644
```

Hospital Size:

```
         SIZE
--------+---------------------
group   |        Total
--------+---------------------
0       |        10304
1       |        14517
2       |        15534
3       |        15895
```

- **After Resampling:**

Age:

```
         AGE
--------+---------------------
group   |        Total
--------+---------------------
60+     |        109027
15-59   |        15918
0-14    |        91784
```

Sex:

```
          SEX
 --------+---------------------
 group   |        Total
 --------+---------------------
 Male    |        138578
 Female  |        78151
```

Race:

```
         RACE
 --------+---------------------
 group   |        Total
 --------+---------------------
 White   |        42615
 Black   |        569
 Others  |        173545
```

Region:

```
          REGION
 --------+---------------------
 group   |        Total
 --------+---------------------
 MA      |        8027
 BA      |        8027
 CE      |        8027
 RN      |        8027
 PR      |        8027
 MG      |        8027
 AC      |        8027
 SP      |        8027
 RR      |        8027
 PA      |        8027
 PI      |        8027
 SC      |        8027
 AL      |        8027
 SE      |        8027
 AM      |        8027
 GO      |        8027
 PE      |        8027
 MS      |        8027
 TO      |        8027
 RO      |        8027
 DF      |        8027
 MT      |        8027
 ES      |        8027
 PB      |        8027
 RS      |        8027
 AP      |        8027
 RJ      |        8027
```

Hospital Size:

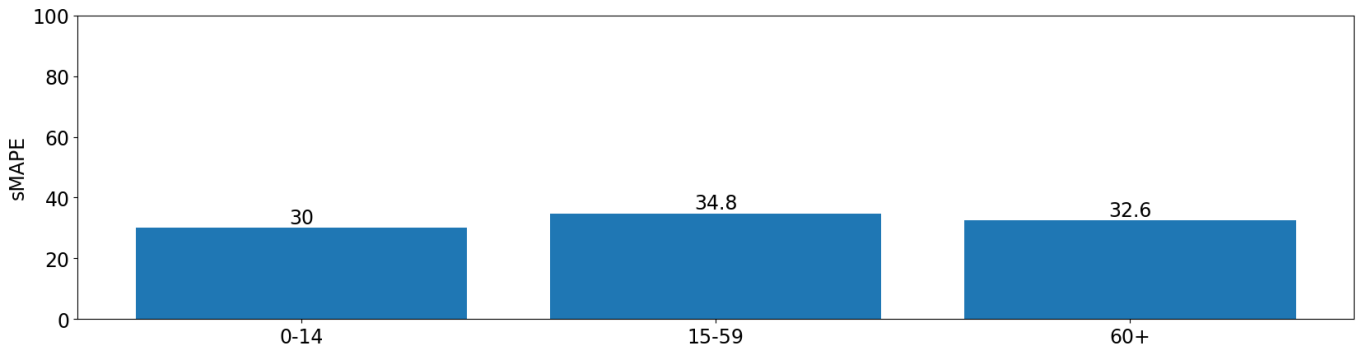
```
         SIZE
--------+--------------------
group   |        Total
--------+--------------------
0       |        39647
1       |        56039
2       |        54346
3       |        66697
```

5: Second Evaluation

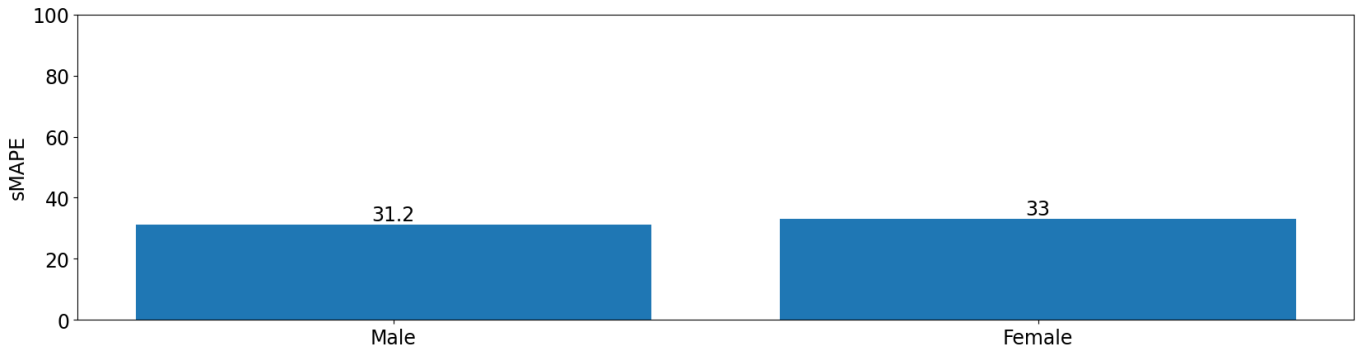
The resampling strategy reduced the max–min regional SMAPE gap from 18.8 to 18.3 percentage points, corresponding to a 2.6% relative reduction. While this indicates small sensitivity to training distribution imbalance, a substantial residual disparity persists. This suggests that regional predictive gaps are not solely attributable to sample imbalance and may reflect structural or data quality asymmetries within the health system. Across other groups, no disparities were created during resampling.

Results:

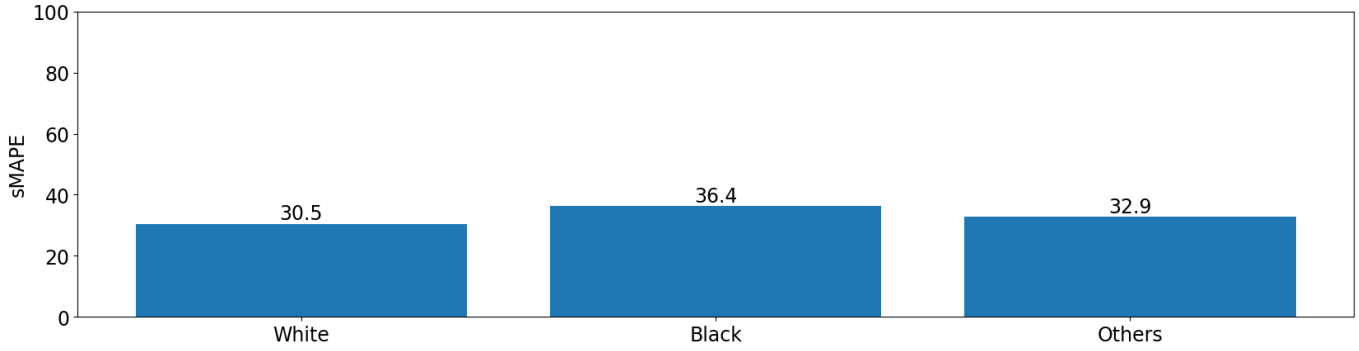
- Age:



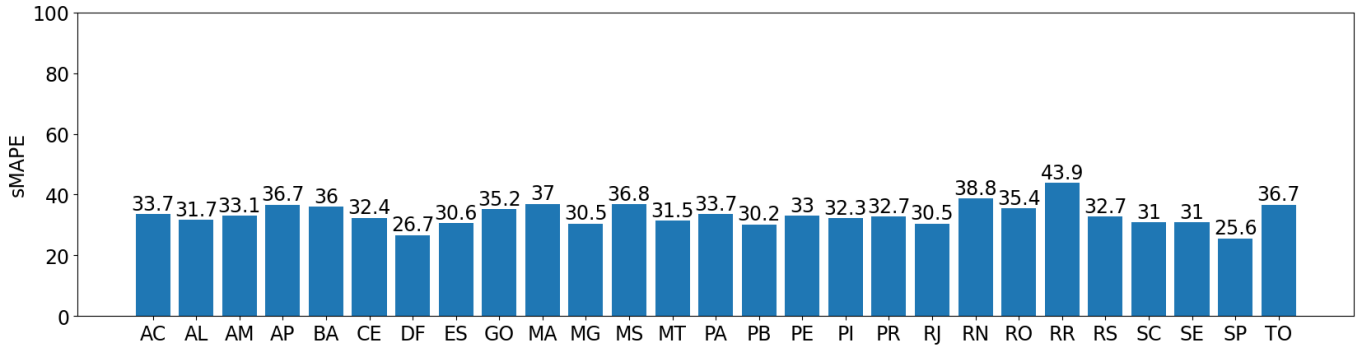
- Sex:



- Race:



- Region:



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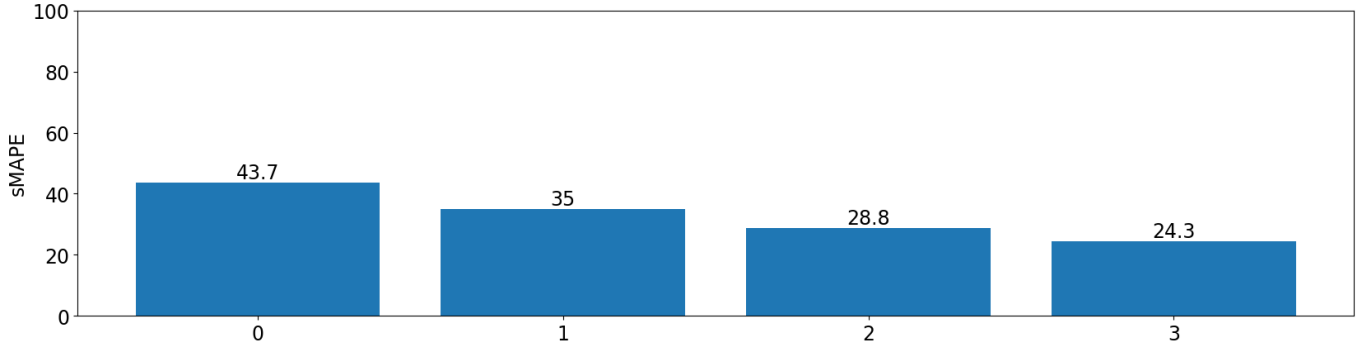

Obs: Many attempts were done at resampling, including test with over-sampling or under-sampling of multiple groups other then region. Many test were also conducted with different seeds. The current finalized version contains the best case scenario found. All other test resulted in worse performance with no gain in bias reduction.